



THE SEVEN ROLES OF AN ENGINEER

Engineering is not one job. Over three weeks you will step into several of these and produce what a professional in that role actually produces.

DESIGN	Turns a requirement into a mechanism, with the calculations to back it. Produces the drawing set or the CAD model.
RESEARCH	Investigates, and predicts a new use for something that exists or a new way to do the job.
DEVELOPMENT	Makes the model or the prototype. Finds out whether the idea survives contact with material.
PRODUCTION & CONSTRUCTION	Works out the order things happen in. Work breakdown structure, then a Gantt chart.
OPERATIONS	Designs the large system and how it is actually run, including what happens when it misbehaves.
SALES	Explains the engineering to someone who has to decide. The ad, the brochure, the pitch.
MANAGEMENT	Builds the team and assigns the work so a product gets finished by people who are not you.